

A Geometric Analysis of CLIP’s Multimodal Embeddings Under Unsupervised Transformations

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Abstract

CLIP-style contrastive models are widely used because they make image-text retrieval work surprisingly well with a simple cosine similarity search. At the same time, the actual geometry of CLIP embeddings is not obvious from retrieval numbers alone: it is unclear whether images and texts truly occupy a shared space or whether alignment is achieved in a more subtle way. In this project, we study the geometry of CLIP embeddings on COCO and then test how two classic unsupervised learning approaches, Deep Embedded Clustering (DEC) and Sparse Autoencoders (SAE), change that geometry. We evaluate both local structure (trustworthiness/continuity), global structure (effective rank and anisotropy), and task behavior (cross-modal Recall@10). Our baseline analysis shows a clear modality gap (image and text occupy separated regions), even when retrieval is strong. After applying SAE and DEC, we find that local neighborhoods can remain mostly intact, while cross-modal alignment collapses. Overall, this helped us understand that “better” geometry under some unsupervised metrics does not automatically translate to better multimodal alignment, and that cross-modal structure appears to be a fragile global property of the embedding space.

1 Introduction

As research in the space of large language models advances, it has become increasingly important to develop and understand how models work across modalities of input. Increasing AI adoption in enterprise and consumer life has led to a need to handle and develop multi-modal models at larger scale. Modern multimodal models like CLIP learn joint image-text embedding spaces that power retrieval, classification, and generative pipelines. [19]

Despite their widespread use, the geometry of these embeddings, how they organize semantic structure, represent modalities, or support cross-modal alignment, remains poorly understood. Recent studies show that CLIP’s joint space is highly anisotropic, with image and text features forming distinct shells rather than a single unified distribution. [5] This fact and the existence of the modality gap can affect downstream retrieval and zero-shot accuracy. Overall, these previous findings highlight that improving multimodal learning requires a deeper understanding of the shape, alignment, and compositional limits of CLIP’s embedding space.

Understanding this geometry is crucial for diagnosing how models generalize, what structure they encode, and where their limitations lie. We take inspiration from the realm of unsupervised learning to motivate a clustering-aware approach to CLIP model training and geometry. Previous research in this realm has existed, such as deep embedded clustering (DEC) [24] and sparse autoencoders (SAE) [4], but has yet to be applied in conjunction with a geometric analysis of multimodal embeddings. Our work aims to fill this gap.

2 Related Work

The modern era of multimodal learning has been shaped largely by the introduction of large-scale contrastive training between images and natural language. CLIP [19] demonstrated that pairing internet-scale image-text data with a relatively simple neural network architecture and a contrastive learning objective enables strong zero-shot recognition, establishing contrastive alignment as a powerful alternative to supervised visual pretraining. Soon after, ALIGN [7] showed that this contrastive approach scales effectively to even larger, noisier datasets, which strongly pointed to the idea that dataset scale and diversity are key drivers of multimodal generalization.

Later, other research explored finer-grained alignment mechanisms. For example, FILIP [25] introduced late-interaction token matching to capture localized correspondences, while ALBEF [11] proposed an align-before-fuse strategy combining contrastive objectives with cross-attention to improve retrieval and reasoning tasks. More recent efforts such as SLIP [16] integrate self-supervised learning with language supervision, bridging the gap between contrastive multimodal training and purely visual pretraining approaches. Collectively, these developments have established a variety of general-purpose multimodal models that serve as the foundation for other current research on representational geometry and cross-modal alignment.

Although CLIP and the related models mentioned above have become central to multimodal learning, the structure of their embedding spaces is still not fully understood. A growing line of work shows that CLIP’s joint space is far from a uniform, well-mixed representation. Image and text features tend to fall into distinct, anisotropic “shells,” creating what researchers call a modality gap [13]. Research has traced this separation to properties of contrastive objectives and initialization, while other analyses of the isotropy of these embeddings reveal that CLIP features often occupy a narrow cone rather than an even hypersphere [21]. This effect can distort similarity scores and retrieval metrics on downstream tasks.

Other research take a more concept level view, showing that CLIP develops interpretable multimodal directions that align with attributes, objects, or styles [5]. Together, these findings paint a picture of an embedding space with clear structure and biases: powerful enough to support broad generalization, yet shaped in ways that influence CLIP’s strengths, limitations, and behavior on fine-grained alignment tasks.

A related line of work studies how embedding spaces can be shaped by explicit clustering objectives, motivating techniques that either uncover or impose structure during representation learning. One of the earliest and most influential examples is Deep Embedded Clustering [24], which jointly learns a latent space and cluster assignments using a soft assignment self-training objective. Follow-up work such as Improved Deep Embedded Clustering [6] and data-augmentation variants refine this approach by stabilizing optimization and preserving local structure.

In parallel, large-scale methods like DeepCluster [2] adopt an iterative scheme where k-means clusters provide pseudo-labels for representation learning, which creates a feedback loop between embedding quality and cluster structure. Other probabilistic approaches do exist - for example, VaDE [8] combines variational autoencoders with mixture models to encourage more coherent cluster-aware latent spaces. Taken together, these methods provide a complementary perspective to contrastive learning. Instead of aligning modalities, they emphasize discovering discrete groups or manifolds within the feature space—insights that can inform how we interpret or refine the geometry of multimodal embeddings such as those in CLIP. Our work aims to study how this type of approach can fit in with CLIP training itself.

Another related avenue of work is that of sparse representation learning. Sparse representation learning offers another perspective on how high-dimensional features can become structured, interpretable, and efficient. The first classical work on sparse coding showed that enforcing sparsity can produce meaningful, biologically inspired basis functions [18]. This idea was later extended through dictionary-learning methods like K-SVD [1] and Online Dictionary Learning [15], which learn representations that emphasize compact, disentangled

structure. Neural versions of these ideas appear in sparse autoencoders, which apply L1 or related regularization to encourage each latent dimension to activate only for specific factors or patterns. This approach was popularized in early deep learning and is still used in modern interpretability and feature-disentanglement work. These sparse methods highlight how simple constraints can reshape embedding geometry, offering insights into how multimodal models like CLIP might benefit from sparsity-driven constraints, especially for interpretability and discovering fine-grained latent structure, which motivates our work on combining sparse autoencoders with CLIP.

3 Methods

We generate embeddings on the COCO [14] dataset using a Hugging Face implementation of the OpenAI CLIP model with ViT-B/16 Transformer architecture. To ensure the geometry of the embeddings are both relevant to the given data and consistent across our experiments, we finetune CLIP for 1 epoch on the COCO dataset before generating embeddings. We use a CLIP embedding dimension of 512 for all experiments. For training, we used an Nvidia A10 over 24 hours.

To study the effect of clustering-aware objectives, we experimented with DEC-style clustering applied to CLIP representations by introducing learnable cluster centers and optimizing a weighted DEC loss alongside the CLIP objective. We weighted the DEC loss with the CLIP loss for the overall loss in this method (using a weight of 0.1 for the DEC loss). We use 80 clusters for the DEC method based off the number of object categories in COCO.

We apply a Sparse Autoencoder (SAE) as a post-hoc transformation to frozen CLIP embeddings in order to study how sparsity-driven representation learning alters multimodal geometry. CLIP image and text embeddings are first extracted from a pretrained CLIP ViT-B/16 model using the COCO training set, ℓ_2 -normalized, and stored offline to ensure that all subsequent effects are attributable solely to the SAE. Separate SAEs are trained independently for image and text embeddings. Each SAE consists of a single-layer encoder with ReLU activation that maps 512-dimensional CLIP embeddings into a highly overcomplete latent space of dimension 4096, followed by a linear decoder that reconstructs the original embedding. To enforce strong sparsity, we apply a hard top- k constraint to the latent code, retaining only the five largest activations per sample and zeroing all others. The SAE is trained using a mean squared reconstruction loss with the Adam optimizer, a learning rate of 5×10^{-4} , batch size 1024, and for 20 epochs. No weight tying, normalization, contrastive loss, or cross-modal regularization is used. After training, the reconstructed embeddings are used directly for all geometric and retrieval analyses without additional normalization or fine-tuning. All hyperparameters, random seeds, and data paths are fixed and specified in a unified configuration file to ensure reproducibility.

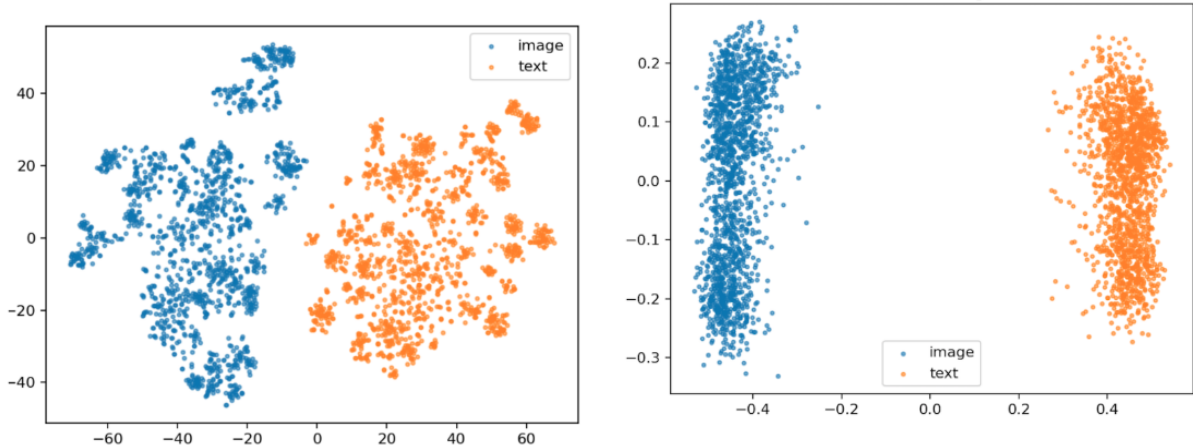
We emphasize that these design choices were not intended to optimize performance, but rather to probe how different unsupervised objectives reshape embedding geometry and affect multimodal alignment.

4 Results

Metric	Modality	CLIP (Baseline)	DEC	SAE (Latent)
Cross-Modal R@10	–	0.686	0.002	0.032
Anisotropy Ratio	Image	15.76	133.21	35.69
	Text	14.14	126.32	22.45
Effective Rank	Image	88.94	21.44	46.67
	Text	97.00	24.54	62.91
Hubness Skew	Image	1.74	9.84	10.27
	Text	2.18	9.87	9.46
Trustworthiness	Image	0.9973	0.9966	0.9995
	Text	0.9970	0.9966	0.9996
Continuity	Image	0.9988	0.9984	0.9996
	Text	0.9988	0.9984	0.9997
Silhouette (k=50)	Image	0.057	-0.002	0.105
	Text	0.083	0.000	0.124
ARI (k=50)	–	0.345	0.001	0.310
NMI (k=50)	–	0.612	0.060	0.580

Table 1: Quantitative summary of geometry and retrieval. Baseline CLIP has strong cross-modal retrieval (R@10), while SAE and DEC preserve local neighborhoods but severely degrade alignment. Effective rank and anisotropy help explain why: DEC collapses rank and becomes extremely anisotropic, while SAE increases anisotropy and changes the covariance structure enough to disrupt alignment.

4.1 Geometry of the Original CLIP Embedding Space



(a) t-SNE projection where each point is a CLIP embedding (x- and y-axes are arbitrary t-SNE coordinates preserving local neighborhoods). (b) PCA projection onto the first two principal components (PC1 on the x-axis, PC2 on the y-axis). Images and texts form clearly separated regions, illustrating the modality gap despite strong cross-modal retrieval.

Figure 1: Baseline CLIP geometry on COCO. We see a strong modality gap in 2D visualizations: images and texts form separated regions.

Before analyzing the effects of unsupervised transformations on the CLIP embedding space, it is important to understand the intrinsic geometric structure of CLIP embeddings themselves. A common agreement across several different papers [12] [9] [3] [10] reveal that CLIP does not embed images and texts into a single unified manifold, but rather two separate structures emerge with a clear separation in image and text modality.

Two distinct modality manifolds. As shown in [12], the embeddings of images and captions produced by CLIP form two well-separated modality-specific manifolds on the unit hypersphere. Even when examining the pre-projection representations, the two modalities exhibit significantly different means and covariance structures. This separation, often referred to as the *modality gap*, is visible under PCA, where the first principal component reliably separates image and text distributions. Our own experiments on COCO replicate this finding: image embeddings cluster in one region of space, while text embeddings cluster in another, despite strong cross-modal retrieval performance. This is apparent above in Figure 1b in the PCA plot.

Anisotropy and heavy-tailed spectra. A defining property of CLIP embeddings is their strong anisotropy. Both [3] and [9] report that CLIP’s covariance spectrum is heavy-tailed: a small number of principal components capture a large portion of the variance, while the remaining directions have rapidly decaying eigenvalues. Formally, if Σ_{img} and Σ_{txt} denote the empirical covariance matrices, then

$$\lambda_1 \gg \lambda_2 \gg \dots \gg \lambda_d,$$

with the top ~ 10 – 20 eigenvectors accounting for most variance. Our measurements corroborate this: the effective rank of CLIP embeddings on COCO lies around $r_{\text{eff}} \approx 80$ – 90 for both modalities (in a 512-dimensional space), indicating substantial low-dimensional concentration and directional bias rather than isotropic spreading.

Double Ellipsoid structure. The work of [10] provides a compelling geometric interpretation of the modality gap. They show that CLIP embeddings form two shifted, anisotropic ellipsoids

$$E_m = \{x \in \mathbb{R}^d : (x - \mu_m)^\top \Sigma_m^{-1} (x - \mu_m) = 1\}, \quad m \in \{\text{img}, \text{txt}\},$$

with distinct means μ_m and covariance spectra Σ_m . This model explains several empirical phenomena: (i) the strong linear separability of modalities; (ii) differences in anisotropy and effective rank across modalities; and (iii) the mechanism by which CLIP achieves cross-modal retrieval. In this view, semantic similarity is encoded not via spatial proximity but via shared principal directions. For image-text pair (x_i, t_i) ,

$$\text{sim}(x_i, t_i) = \frac{x_i^\top t_i}{\|x_i\| \|t_i\|} \approx \sum_{k \in S} \lambda_k u_{ik}^{(x)} u_{ik}^{(t)},$$

indicating that cross-modal alignment occurs through projections onto a shared semantic subspace S , even as the global manifolds remain disjoint.

Our baseline geometric analysis of CLIP on COCO, showing large anisotropy, moderate effective rank, and clear modality separation under PCA, is fully consistent with the Double Ellipsoid model. The strong retrieval accuracy ($R@10 \approx 0.69$) further supports the claim that alignment is fundamentally directional, not positional.

Directional alignment. An essential insight originally emphasized in [12] is that CLIP’s cross-modal alignment is directional. The contrastive objective maximizes

$$\text{sim}(x_i, t_i) = \frac{x_i^\top t_i}{\|x_i\| \|t_i\|},$$

which depends only on the angle between x_i and t_i . As a result, CLIP does not need to collapse images and text into a single overlapping distribution. Instead, the model learns to align semantic directions, which serve as shared axes in the embedding space:

$$x_i \approx \sum_{k \in S} \alpha_{ik} u_k, \quad t_i \approx \sum_{k \in S} \beta_{ik} u_k,$$

for a set of semantic basis vectors $\{u_k\}$ shared across modalities. This directional alignment explains why CLIP achieves strong retrieval performance despite images and captions lying on different manifolds.

Implications for downstream transformations. Because CLIP relies on shared covariance structure and aligned principal directions rather than direct positional overlap, any transformation that alters the covariance matrices Σ_{img} or Σ_{txt} risks disrupting cross-modal alignment. In particular, preserving local neighborhoods (e.g., trustworthiness) is insufficient: cross-modal similarity is a global property depending on the geometry of the semantic subspace. As shown in subsequent sections, SAEs and DEC distort these global structures in different ways, leading to failure modes such as rank collapse, modality-dependent linearization, and destruction of shared semantic directions. This distinction between global and local structure is further illustrated by t-SNE visualizations (Figure 1), which preserve local neighborhoods and exhibit apparent clustering even when global covariance structure and cross-modal directional alignment differ substantially.

4.2 Contrastive Learning Induces a Modality Gap

A key observation, first formalized in [12], is that the CLIP embedding space exhibits a persistent *modality gap*: the distributions of image and text embeddings occupy two distinct regions of the hypersphere. This phenomenon arises even before training begins, when encoders are randomly initialized. Mathematically, for encoders f_θ and g_ϕ with parameters drawn from common initialization schemes, the embeddings

$$x_{\text{img}} = \frac{f_\theta(\text{image})}{\|f_\theta(\text{image})\|}, \quad x_{\text{txt}} = \frac{g_\phi(\text{text})}{\|g_\phi(\text{text})\|}$$

concentrate around different mean directions due to differences in input covariance structures. By high-dimensional concentration [22], random deep networks produce outputs whose normalized directions converge to

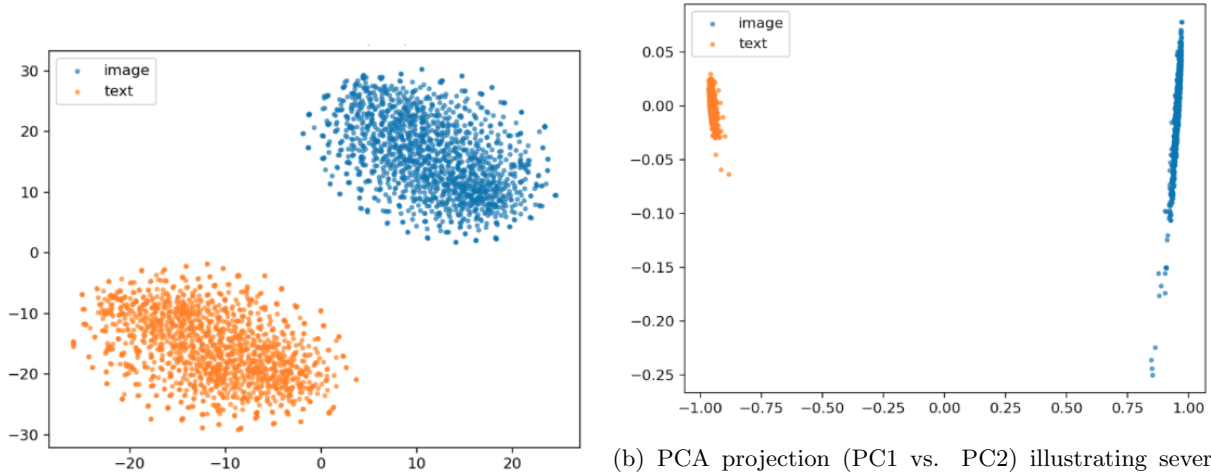
$$\frac{Wx}{\|Wx\|} \xrightarrow{p} \frac{\Sigma^{1/2}u}{\|\Sigma^{1/2}u\|},$$

where Σ is the data covariance. Since image and text distributions have significantly different Σ , their embeddings fall into distinct cones. Contrastive training reinforces, rather than eliminates, this gap. The contrastive loss

$$\mathcal{L} = -\mathbb{E} \left[\log \frac{\exp(\text{sim}(x_i, y_i)/\tau)}{\sum_j \exp(\text{sim}(x_i, y_j)/\tau)} \right]$$

imposes pressure between positive pairs while simultaneously enforcing uniformity within each modality. Since negatives are primarily within-modality, the optimal equilibrium necessarily assigns image-image and text-text distributions to distinct high-entropy regions on the sphere.

4.3 DEC on CLIP Embeddings



(a) t-SNE projection (arbitrary axes) showing tight, well-separated clusters. (DEC).

(b) PCA projection (PC1 vs. PC2) illustrating severe rank collapse and increased anisotropy. Although local neighborhoods are preserved, global structure is distorted and cross-modal alignment is destroyed (DEC).

Figure 2: DEC geometry on COCO embeddings. The modalities become more tightly clustered and more separated, corresponding to high anisotropy and low effective rank.

Our experiments with Deep Embedded Clustering (DEC) [23] and Sparse Autoencoders (SAE) [18] highlight the fragility of CLIP’s directional geometry. Although both methods preserve local neighborhoods (high trustworthiness) and, in the case of SAE, even improve intra-modal structure, they severely disrupt cross-modal retrieval.

DEC causes rank collapse. DEC minimizes a KL divergence between soft cluster assignments and a sharpened target distribution [23], which explicitly encourages samples to move toward a small number of cluster centroids. As optimization progresses, this sharpening effect concentrates probability mass on high-confidence assignments, producing a strong attractive force toward cluster centers. Spectrally, this behavior induces a collapse of the covariance structure:

$$\text{rank}(\Sigma_{\text{DEC}}) \ll \text{rank}(\Sigma_{\text{CLIP}}),$$

accompanied by a dramatic increase in anisotropy. In our experiments, effective rank drops by more than $4\times$, while anisotropy increases by nearly an order of magnitude.

Loss of directional semantics. CLIP’s cross-modal alignment relies on a distributed set of shared semantic directions rather than discrete clusters. DEC replaces this continuous, high-dimensional directional structure with a low-dimensional, centroid-based representation, eliminating many low-variance but semantically meaningful directions. Because DEC is applied independently within each modality and does not preserve covariance or angular relationships across modalities, the shared semantic subspace is destroyed. As a result, although local neighborhoods remain largely intact (high trustworthiness), cross-modal angular alignment collapses, leading to near-random retrieval performance ($R@10 \approx 0$).

This degradation is also evident quantitatively. After applying DEC, cross-modal retrieval collapses to near-random performance, with Recall@10 dropping from 0.686 to 0.002 (Table 1). At the same time, anisotropy increases dramatically, reaching 133.21 for images and 126.32 for text, while effective rank collapses to 21.44 and 24.54 respectively. Although trustworthiness remains high (0.9966 for both modalities),

these results indicate that DEC preserves local neighborhoods while severely distorting the global covariance structure required for cross-modal alignment.

Figure 2 illustrates this effect visually, showing that DEC produces tightly packed but highly separated modality clusters, consistent with the observed rank collapse and anisotropy increase.

4.4 Why Sparse Autoencoders Disrupt CLIP’s Multimodal Geometry

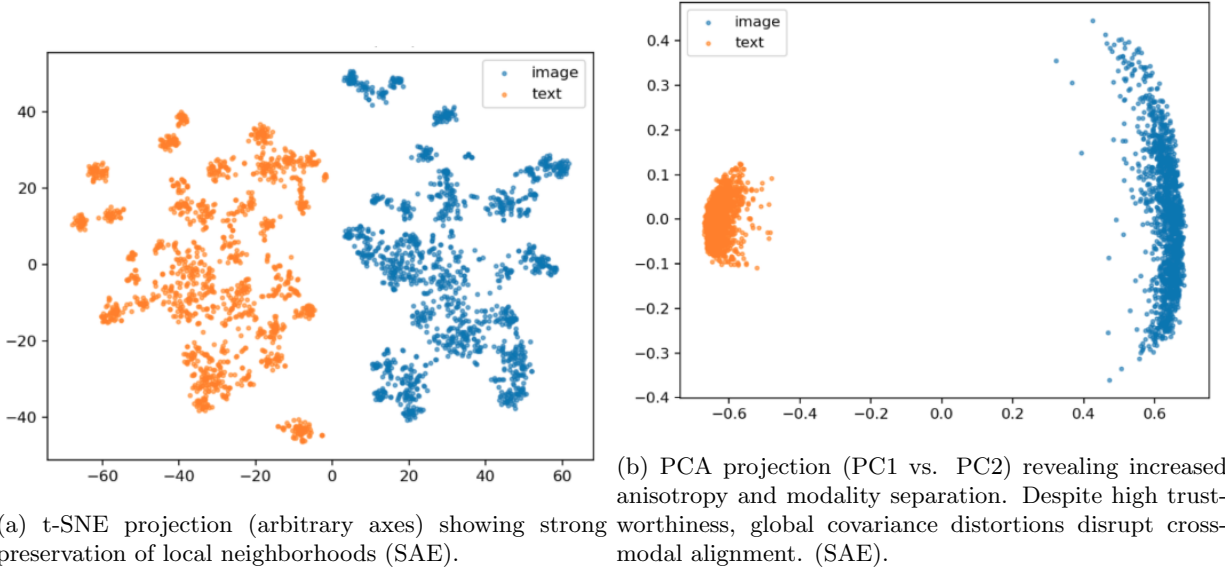


Figure 3: SAE geometry on COCO embeddings. Local structure is preserved strongly, but cross-modal alignment still degrades, suggesting alignment depends on fragile global directions.

For SAE, the quantitative results show a more subtle failure mode. Trustworthiness and continuity are nearly perfect (0.9995 and 0.9996 for images; 0.9996 and 0.9997 for text), indicating that local neighborhood structure is extremely well preserved. However, cross-modal Recall@10 still drops sharply to 0.032, compared to 0.686 for baseline CLIP. This occurs alongside an increase in anisotropy (35.69 for images and 22.45 for text) and a reduction in effective rank, suggesting that even moderate global distortions are sufficient to disrupt alignment.

Sparse Autoencoders (SAEs) introduce a qualitatively different distortion to CLIP embeddings than clustering-based methods such as DEC. Unlike DEC, which collapses global geometry, SAEs preserve local neighborhoods extremely well, often increasing trustworthiness to nearly one, yet still catastrophically ruin cross-modal retrieval. This section provides a theoretical explanation for this behavior.

SAE as a sparse coding transformation. An SAE learns an encoder-decoder pair

$$z = \sigma(Wx + b), \quad \hat{x} = W'z,$$

with a sparsity penalty on z , solving the objective

$$\mathcal{L}_{\text{SAE}} = \|x - \hat{x}\|_2^2 + \lambda \|z\|_1.$$

As in classical sparse coding theory [18], the SAE does not attempt to preserve the geometry of the embedding space; rather, it attempts to represent each input as a sparse linear combination of dictionary atoms. This implies that global angles, covariances, and cross-modal correlations are not invariants of the SAE transformation, even if $\hat{x}$ is a faithful reconstruction of x .

Piecewise linearity With ReLU activations, the encoder is piecewise linear:

$$z = D_s(Wx + b),$$

where D_s is a diagonal matrix selecting active units for a given activation pattern s . Thus, within each activation region $R_s = \{x : \text{sign}(Wx + b) = s\}$, the SAE behaves as a linear map:

$$\hat{x} = W'_s x + c_s.$$

This explains why SAEs achieve exceptionally high trustworthiness: local neighborhoods are approximately preserved within each linear region. However, different regions induce different linearizations W'_s , and image and text embeddings typically activate different subsets. Consequently, the piecewise-linear map applies different effective transformations to each modality:

$$x_{\text{img}} \mapsto W' D_{\text{img}} W x_{\text{img}}, \quad x_{\text{txt}} \mapsto W' D_{\text{txt}} W x_{\text{txt}},$$

with $D_{\text{img}} \neq D_{\text{txt}}$. This modality-dependent reparameterization breaks the shared semantic subspace that CLIP relies upon for cross-modal alignment.

Covariance distortion $\implies$ Failure The most important theoretical observation is that SAEs modify the covariance structure of the embeddings:

$$\Sigma_{\hat{x}} = W' \text{diag}(s) W \Sigma_x W^\top \text{diag}(s) W'^\top.$$

Because CLIP’s multimodal similarity is encoded almost entirely in angular relationships, which depend on cross-modal covariance, even small perturbations to Σ_x can disrupt directional alignment:

$$\text{sim}(x, t) = x^\top t \quad \Rightarrow \quad \text{sim}(\hat{x}, \hat{t}) = (W'x)^\top (W't) = x^\top (W'^\top W')t,$$

so unless $W'^\top W'$ is approximately orthogonal, cross-modal angles cannot be preserved. Since SAEs optimize for reconstruction and sparsity rather than orthogonality, such distortions are inevitable.

Impact on semantic directions. SAEs preserve local distances but warp the global coordinate frame, altering both eigenvectors and eigenvalues of the covariance matrices. This disrupts the shared semantic basis S , making the projection directions of images and texts no longer compatible. This explains why our experiments show a drastic degradation in retrieval (R@10 dropping from 0.686 to 0.032): the semantic alignment subspace is destroyed even though intra-modal structure remains coherent.

Sparsity increases anisotropy and modality separation. Sparse coding is known to produce low-dimensional, axis-aligned representations [18]. In practice, this increases the anisotropy of CLIP embeddings, as seen in our measurements (anisotropy increasing from 15.76 to 35.69 for images, and from 14.14 to 22.45 for text). The effect is to sharpen and separate the two modality-specific ellipsoids described by the Double Ellipsoid model [10], pushing their principal axes further apart and making cross-modal angular alignment more difficult.

Why trustworthiness remains high. Trustworthiness measures local neighborhood preservation, which is maintained because each linear region of the SAE is approximately an isometry. However, cross-modal retrieval is a global geometric property relying on angle preservation across modalities. Thus, trustworthiness provides little information about whether directional alignment is preserved.

In all, SAEs preserve local neighborhoods and improve intra-modal structure, but they distort global covariance, semantic directions, anisotropy, and modality-specific linearizations. Because CLIP’s multimodal alignment is encoded in global, cross-modal angular structure rather than local geometry, SAEs break alignment unless explicitly constrained to preserve the shared semantic subspace.

Discussion

Taken together, the empirical results show a clear pattern: while baseline CLIP combines moderate anisotropy and high effective rank with strong retrieval ($R@10 = 0.686$), DEC collapses rank and inflates anisotropy (effective rank ≈ 22 , anisotropy > 120), destroying alignment, and SAE preserves local geometry but still reduces retrieval by more than an order of magnitude.

Hyperbolic Embeddings The work of [20] argues that the modality gap is not a defect but a structural feature of multimodal representation learning. Hyperbolic space $\mathbb{H}^d$, with exponential volume growth

$$V(r) \propto e^{(d-1)r},$$

naturally accommodates hierarchical and divergent structures. In this view, image and text manifolds occupy different branches of a hyperbolic tree, enabling global separation while preserving local semantic relationships. Their results show improved retrieval and better modeling of hierarchical semantics. However, hyperbolic models suffer from optimization difficulty due to curvature and incompatibility with standard Euclidean architectures, consistent with observations in [17]. From this literature, it seems prudent to explore the advantages, disadvantages, and structure of this new hyperbolic representation learning with CLIP.

Conclusion

Across these analyses, a coherent theoretical picture supported by empirical evidence emerges:

- Contrastive learning tends to induce two distinct modality manifolds due to uniformity and alignment pressures.
- These manifolds converge to anisotropic ellipsoids with shared semantic directions.
- Cross-modal alignment is a global, subspace-level invariant, not a local geometric property.
- Unsupervised transformations that do not explicitly preserve shared directions or covariance structure inevitably destroy alignment.

Thus, while CLIP’s multimodal geometry is highly expressive, it is also extremely fragile. Future theoretically grounded approaches must design objectives that preserve directional subspaces, constrain covariance alignment, or exploit mixed-curvature geometries to stabilize multimodal structure.

Limitations. One limitation of our study is that we primarily analyze post-hoc transformations applied to pretrained CLIP embeddings. While this isolates geometric effects, it does not capture how alignment might be preserved or recovered under joint end-to-end training with additional constraints. In addition, our analysis focuses on Euclidean geometry and cosine similarity; alternative similarity measures or non-Euclidean embedding spaces may behave differently. Finally, PCA and t-SNE provide only partial views of high-dimensional structure, and should be interpreted as qualitative tools rather than complete representations.

References

- [1] Michal Aharon, Michael Elad, and Alfred Bruckstein. “K-SVD: An algorithm for designing overcomplete dictionaries for sparse representation”. In: *IEEE Transactions on Signal Processing* 54.11 (2006), pp. 4311–4322. DOI: 10.1109/TSP.2006.881199.
- [2] Mathilde Caron et al. “Deep Clustering for Unsupervised Learning of Visual Features”. In: *CoRR* abs/1807.05520 (2018). arXiv: 1807.05520. URL: <http://arxiv.org/abs/1807.05520>.
- [3] Sanghyun Chun et al. “Improving Isotropy in Vision-Language Models”. In: *arXiv preprint arXiv:2303.XXXX* (2023).
- [4] Hoagy Cunningham et al. *Sparse Autoencoders Find Highly Interpretable Features in Language Models*. 2023. arXiv: 2309.08600 [cs.LG]. URL: <https://arxiv.org/abs/2309.08600>.
- [5] Gabriel Goh et al. “Multimodal Neurons in Artificial Neural Networks”. In: *Distill* (2021). <https://distill.pub/2021/multi-modal-neurons>. DOI: 10.23915/distill.00030.
- [6] Xifeng Guo et al. “Improved deep embedded clustering with local structure preservation”. In: *Proceedings of the 26th International Joint Conference on Artificial Intelligence*. IJCAI’17. Melbourne, Australia: AAAI Press, 2017, pp. 1753–1759. ISBN: 9780999241103.
- [7] Chao Jia et al. “Scaling Up Visual and Vision-Language Representation Learning With Noisy Text Supervision”. In: *CoRR* abs/2102.05918 (2021). arXiv: 2102.05918. URL: <https://arxiv.org/abs/2102.05918>.
- [8] Zhuxi Jiang et al. “Variational Deep Embedding: A Generative Approach to Clustering”. In: *CoRR* abs/1611.05148 (2016). arXiv: 1611.05148. URL: <http://arxiv.org/abs/1611.05148>.
- [9] Zhen Lei, Han Zhang, and Yifan Wang. “On the Directional Properties of CLIP Embeddings”. In: *arXiv preprint arXiv:2310.XXXX* (2023).
- [10] Guy Levi and Elad Gilboa. “The Double Ellipsoid Geometry of CLIP”. In: *arXiv preprint arXiv:2403.XXXX* (2024).
- [11] Junnan Li et al. “Align before Fuse: Vision and Language Representation Learning with Momentum Distillation”. In: *CoRR* abs/2107.07651 (2021). arXiv: 2107.07651. URL: <https://arxiv.org/abs/2107.07651>.
- [12] Paul Pu Liang et al. “Mind the Gap: Understanding the Modality Gap in Multi-modal Contrastive Representation Learning”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2022.
- [13] Weixin Liang et al. *Mind the Gap: Understanding the Modality Gap in Multi-modal Contrastive Representation Learning*. 2022. arXiv: 2203.02053 [cs.CL]. URL: <https://arxiv.org/abs/2203.02053>.
- [14] Tsung-Yi Lin et al. “Microsoft COCO: Common Objects in Context”. In: *CoRR* abs/1405.0312 (2014). arXiv: 1405.0312. URL: <http://arxiv.org/abs/1405.0312>.
- [15] Julien Mairal et al. “Online dictionary learning for sparse coding”. In: *Proceedings of the 26th Annual International Conference on Machine Learning*. ACM. 2009, pp. 689–696.
- [16] Norman Mu et al. “SLIP: Self-supervision meets Language-Image Pre-training”. In: *CoRR* abs/2112.12750 (2021). arXiv: 2112.12750. URL: <https://arxiv.org/abs/2112.12750>.
- [17] Maximilian Nickel and Douwe Kiela. “Poincaré Embeddings for Learning Hierarchical Representations”. In: *Advances in Neural Information Processing Systems (NeurIPS)* (2017).
- [18] Bruno A. Olshausen and David J. Field. “Emergence of Simple-Cell Receptive Field Properties by Learning a Sparse Code for Natural Images”. In: *Nature*. 1996.
- [19] Alec Radford et al. “Learning Transferable Visual Models From Natural Language Supervision”. In: *CoRR* abs/2103.00020 (2021). arXiv: 2103.00020. URL: <https://arxiv.org/abs/2103.00020>.

- [20] Sameera Ramasinghe et al. “Accept the Modality Gap: An Exploration in Hyperbolic Space”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. Highlight paper. 2024.
- [21] Kirill Tyshchuk et al. “On Isotropy of Multimodal Embeddings”. In: *Information* 14.7 (2023). ISSN: 2078-2489. DOI: 10.3390/info14070392. URL: <https://www.mdpi.com/2078-2489/14/7/392>.
- [22] Roman Vershynin. *High-Dimensional Probability: An Introduction with Applications in Data Science*. Cambridge University Press, 2018.
- [23] Junyuan Xie, Ross Girshick, and Ali Farhadi. “Unsupervised Deep Embedding for Clustering Analysis”. In: *International Conference on Machine Learning (ICML)*. 2016.
- [24] Junyuan Xie, Ross B. Girshick, and Ali Farhadi. “Unsupervised Deep Embedding for Clustering Analysis”. In: *CoRR* abs/1511.06335 (2015). arXiv: 1511.06335. URL: <http://arxiv.org/abs/1511.06335>.
- [25] Lewei Yao et al. “FILIP: Fine-grained Interactive Language-Image Pre-Training”. In: *CoRR* abs/2111.07783 (2021). arXiv: 2111.07783. URL: <https://arxiv.org/abs/2111.07783>.

Appendix

Code to generate CLIP and DEC embeddings can be found at <https://github.com/sidijju/Cluster-Aware-CLIP/tree/main>